PretoxTM: Preclinical Text Mining system for the extraction of treatment-related findings from toxicology study reports

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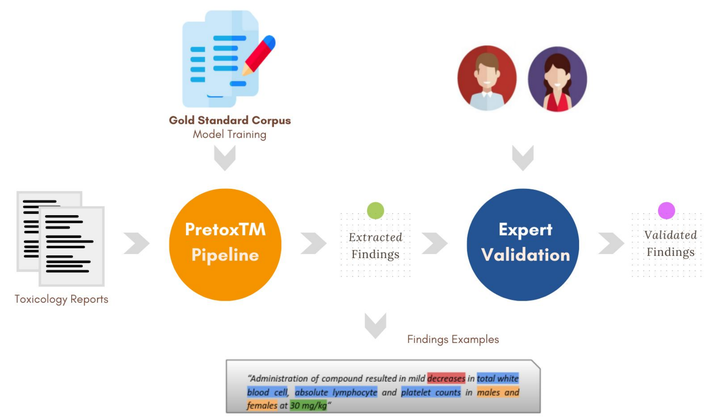
Abstract

Over the last few decades the pharmaceutical industry has generated a vast corpus of knowledge on the safety and efficacy of drugs. Much of this information is contained in **toxicology reports**, which describe a series of animal studies designed to analyse the effects of the tested compound, including unintended pharmacological and toxic effects, known as **treatment-related findings**. Despite the potential of this knowledge, the fact that most of this relevant information is only available as unstructured text with variable degrees of digitisation has hampered its systematic access, use and exploitation. In the context of the **eTRANSAFE** project, we present PretoxTM (Preclinical Toxicology Text Mining), the first system specifically designed to detect, extract, organise and visualise treatment-related findings from toxicology reports. The PretoxTM tool comprises three main components: **PretoxTM Corpus**, **PretoxTM Pipeline** and **PretoxTM Web App**. The PretoxTM Corpus is a gold standard corpus of preclinical treatment-related findings annotated by toxicology experts. This corpus was used to develop, train and validate the PretoxTM Pipeline that extracts treatment-related findings using a fine-tuned Transformer model. The extracted information is then presented for expert visualization and validation in the PretoxTM Web App.

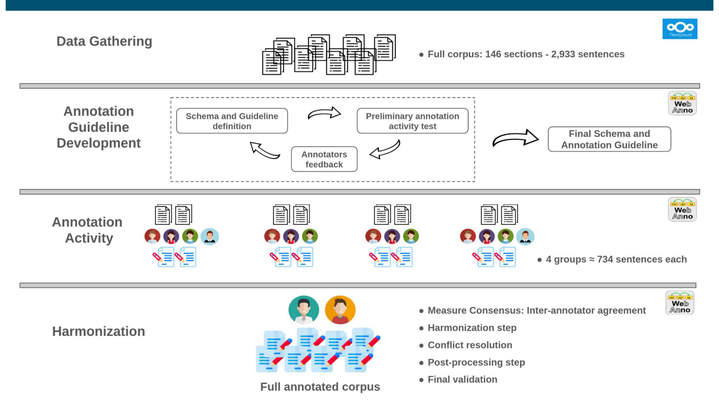
PretoxTM Overview



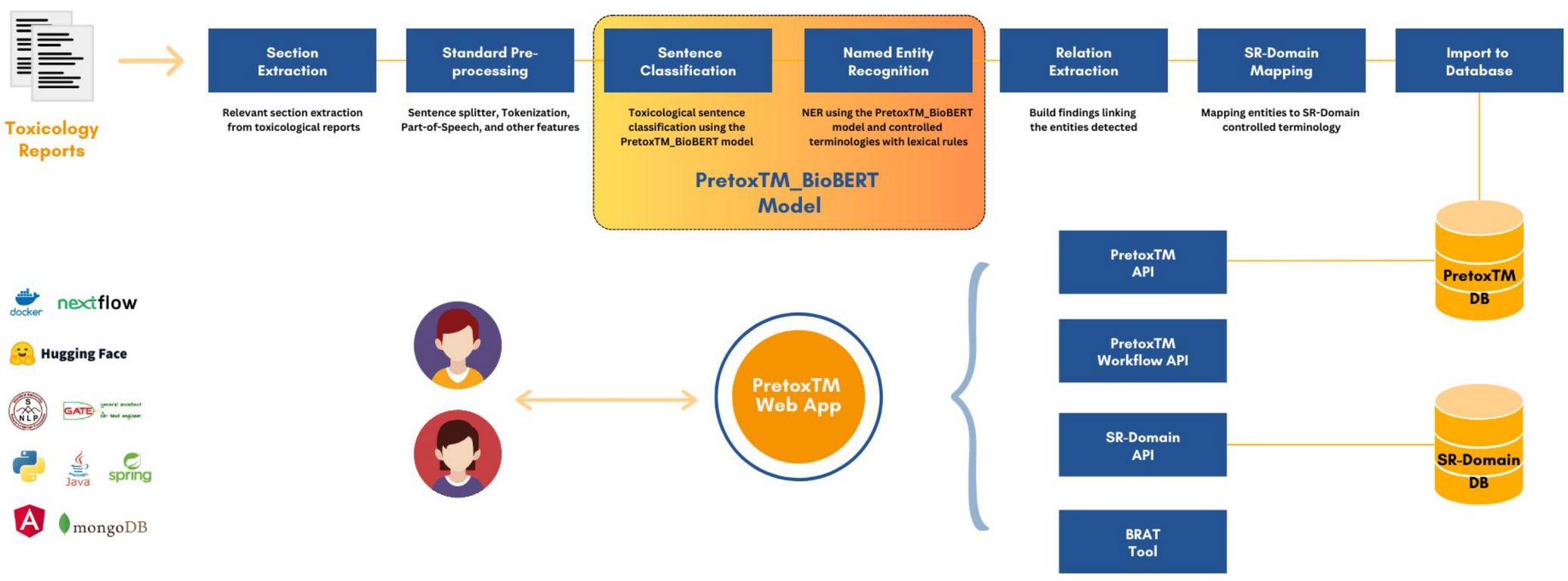
Treatment-related findings

A **treatment-related finding expression** encloses several named entities. The most relevant one is the abnormal effect detected, which, depending on the study domain of the finding, can be given by a measurement, a test, or an examination named **Study Test** and an abnormal **Manifestation** result obtained for that study test; or by an abnormal **Finding** in study domains where there is no associated test or measurement (e.g. clinical, macroscopic and microscopic). Other related named entities that could be present to complete the treatment-related finding are: the **Specimen** of the abnormal observation, the **Sex** of the subject, the **Group** of subjects in which the observation was detected and the **Dose** level administration of the compound. Examples of sentences with treatment-related findings are: "*The decrease in food consumption and body weight of the animals from the mid dose onwards is regarded as evidence of general toxicity.*" and "*At dose level 3, absolute and relative liver weights were increased in male rats.*".

PretoxTM Corpus development



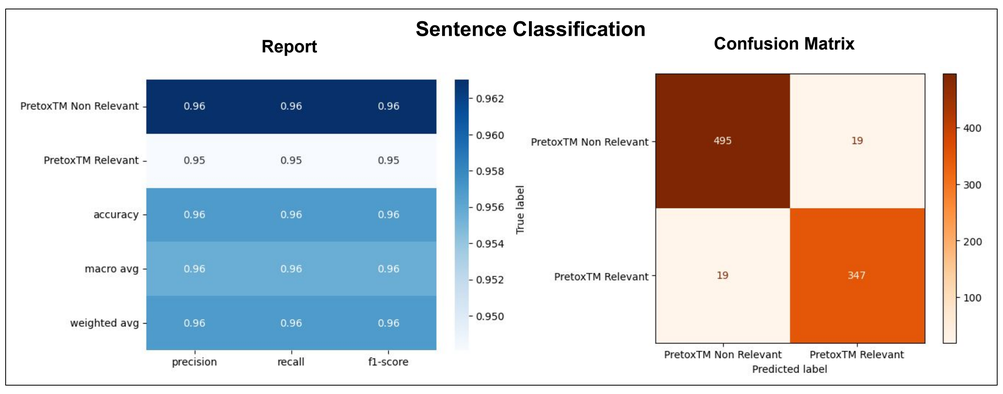
PretoxTM System Architecture

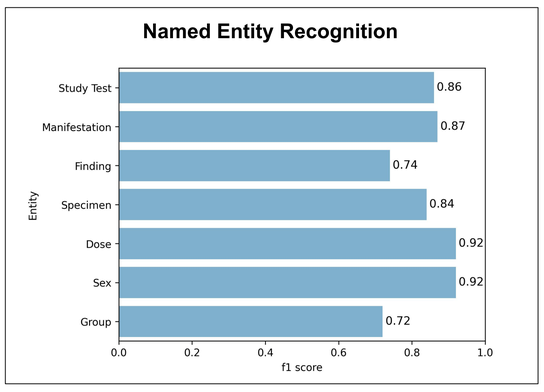


The **PretoxTM pipeline** is responsible for the extraction of treatment-related findings from toxicology reports. The PretoxTM pipeline was developed using various NLP techniques, among which we highlight the use of **Transformers**. We developed the **PretoxTM_BioBERT** model, in which we have performed a fine-tuning of the BioBERT pre-trained model using the PretoxTM Corpus for Sentence Classification and Named Entity Recognition (NER). BioBERT is a domain-specific language representation model pre-trained on large-scale biomedical corpora. After the execution of the pipeline, the extracted findings are deposited in the **PretoxTM Database**.

The **PretoxTM Web App resources** comprise the collection of components developed in order to provide a user interface for toxicology expert validation and programmatic access to the PretoxTM Database. The **PretoxTM Web App** provides the following functionality: upload study reports, run the PretoxTM pipeline, visualise and validate the extracted findings.

PretoxTM NLP Performance





PretoxTM Web App

https://pretoxtm.gitlab.io/documentation/#user-manual-video

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